

Problem A is the finite Bessel-energy class

A literature-implied full characterization for arXiv:1412.5855

Run fa_banach_001 — agent_lane_03

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Status. Full answer to the stated characterization problem, but not an originality claim. It is a direct specialization of classical Bessel-potential theory.

The source problem

Jamróz writes $\mathcal{M}_+(\mathbb{R}^2)$ for the finite nonnegative Radon measures and defines $H^{-1}(\mathbb{R}^2)$ by the Fourier weight $(1 + |\xi|^2)^{-1}$. Problem A asks for a characterization of their intersection. The source then quotes a positive-logarithmic-energy criterion under a compact-support hypothesis.

We comment on possible numerical applications.

Keywords: embeddings of measures, vorticity, Hausdorff dimension

MSC 2010: 46E27, 46E30, 46E35, 28A78

1 Introduction

Let $\mathcal{M}_+(\mathbb{R}^2)$ denote the space of nonnegative bounded Radon measures on $\mathbb{R}^2$ (see [7]) and let $H^{-1}(\mathbb{R}^2)$ be the space of all tempered distributions f on $\mathbb{R}^2$ such that

$$\int_{\mathbb{R}^2} (1 + |y|^2)^{-1} |\hat{f}(y)|^2 dy < \infty.$$

Alternatively, $H^{-1}(\mathbb{R}^2)$ can be viewed as the space of all continuous functionals on the Sobolev space $W^{1,2}(\mathbb{R}^2)$ (see e.g. [1]). The following basic problem can be posed:

Problem A. Characterize the space $\mathcal{M}_+(\mathbb{R}^2) \cap H^{-1}(\mathbb{R}^2)$.

Our motivation to study this problem originates in fluid mechanics. Namely, let $u : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the velocity field of a fluid in two-dimensional space and let

$$\omega = \text{curl}(u) := \partial_{x_1} u_2 - \partial_{x_2} u_1$$

be its vorticity field. Then $\omega \in \mathcal{M}_+(\mathbb{R}^2) \cap H^{-1}(\mathbb{R}^2)$ for compactly supported ω means that

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The exact source pages are bundled as `source_excerpt_problem_A_pages_1_2.pdf`. For a finite nonnegative measure μ , put

$$\mathcal{H}^+(\mu) = \iint_{\mathbb{R}^2 \times \mathbb{R}^2} \log^+ \frac{1}{|x - y|} d\mu(x) d\mu(y).$$

Theorem 1 (Global characterization). *For every finite nonnegative Radon measure μ on $\mathbb{R}^2$,*

$$\mu \in H^{-1}(\mathbb{R}^2) \iff \mathcal{H}^+(\mu) < \infty.$$

Thus the same criterion characterizes all of $\mathcal{M}_+(\mathbb{R}^2) \cap H^{-1}(\mathbb{R}^2)$; compact support is unnecessary.

Proof

Use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} e^{-ix \cdot \xi} f(x) dx, \quad f(x) = (2\pi)^{-2} \int_{\mathbb{R}^2} e^{ix \cdot \xi} \widehat{f}(\xi) d\xi.$$

Let

$$h_t(x) = \frac{1}{4\pi t} e^{-|x|^2/(4t)}, \quad G(x) = \int_0^\infty e^{-t} h_t(x) dt.$$

Then $\widehat{h}_t(\xi) = e^{-t|\xi|^2}$ and hence

$$\widehat{G}(\xi) = \int_0^\infty e^{-t(1+|\xi|^2)} dt = \frac{1}{1+|\xi|^2}.$$

For fixed $t > 0$, Fourier inversion for the bounded heat kernel gives

$$\iint h_t(x-y) d\mu(x) d\mu(y) = (2\pi)^{-2} \int_{\mathbb{R}^2} e^{-t|\xi|^2} |\widehat{\mu}(\xi)|^2 d\xi.$$

All quantities are nonnegative. Tonelli's theorem therefore yields the extended-real identity

$$\iint G(x-y) d\mu(x) d\mu(y) = (2\pi)^{-2} \int_{\mathbb{R}^2} \frac{|\widehat{\mu}(\xi)|^2}{1+|\xi|^2} d\xi. \quad (1)$$

In particular, $\mu \in H^{-1}(\mathbb{R}^2)$ exactly when its G -energy is finite.

The heat integral is the order-two Bessel kernel in dimension two:

$$G(x) = \frac{1}{2\pi} K_0(|x|),$$

where K_0 is the modified Bessel function. The classical asymptotics are

$$K_0(r) \sim \log \frac{1}{r} \quad (r \downarrow 0), \quad K_0(r) \sim \sqrt{\frac{\pi}{2r}} e^{-r} \quad (r \rightarrow \infty).$$

Consequently there is a constant C such that, for every finite positive μ of mass m ,

$$\mathcal{H}^+(\mu) \leq C \iint G(x-y) d\mu(x) d\mu(y),$$

and

$$\iint G(x-y) d\mu(x) d\mu(y) \leq C(\mathcal{H}^+(\mu) + m^2).$$

Indeed, on $0 < r < 1$ the two singular kernels are comparable, while on $r \geq 1$ the Bessel kernel is bounded. Finiteness of the two energies is therefore equivalent. Combining this with (1) proves the theorem. $\square$

Proof Intuition

The inhomogeneous Sobolev weight is already a positive-definite interaction kernel. Writing its reciprocal as a Laplace transform of heat multipliers turns the Fourier norm into a manifestly positive pair energy. In two dimensions that interaction is K_0 : logarithmic at short range and rapidly decaying at long range. A finite measure has no difficulty at long range, so only the positive logarithmic self-interaction decides membership.

Literature relation, scope, and review recommendation

Aronszajn and Smith give the Bessel kernel’s Fourier multiplier and explicit modified-Bessel formula on printed pp. 413–414, and the K_0 asymptotics on printed p. 415. Those pages are bundled in `supporting_excerpt_bessel_kernel_printed_413_415.pdf`. The theorem above is therefore a direct two-dimensional finite-measure specialization of their framework, not a priority claim.

The conclusion fully answers Problem A as phrased. It does not solve the separate Euler uniqueness problem mentioned as motivation. Human review can focus on the extended-real identity (1), the source’s meaning of “bounded Radon measure” as finite mass, and the two kernel comparisons.

References

1. G. Jamróz, *Nonnegative measures belonging to $H^{-1}(\mathbb{R}^2)$* , arXiv:1412.5855, Problem A on PDF p. 1 and Lemma 1.1 on PDF p. 2.
2. N. Aronszajn and K. T. Smith, *Theory of Bessel Potentials. I*, Ann. Inst. Fourier **11** (1961), 385–475; formulas (2.7)–(2.10), (3.4), and (3.5), printed pp. 414–415.